ΕΘΝΙΚΟΝ & ΚΑΠΟΔΙΣΤΡΙΑΚΟΝ ΠΑΝΕΠΙΣΤΗΜΙΟΝ ΑΘΗΝΩΝ
NATIONAL & KAPODISTRIAN UNIVERSITY OF ATHENS

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On the Similarity of (Spatial, Temporal, and) Spatio-Temporal Datases

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NSF

MODAP
mobility, data mining, and privacy

Hurricanes in Texas and Florida; Fires in California and Wyoming

- What does it mean to say that they have *similar motions*?

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Outline

- **Introduction and Motivation**
  - Application Scenarios
    - Semantics
- **Similarity of Spatial/Geometric Entities**
  - Rigid Transformations
- **Similarity and Distance Functions**
  - Properties
- **Similarity and Time-Series Data**
  - Time Series and Trajectories
- **Similarity of Moving Objects Trajectories**
  - Models/Querying/Indexing
  - Patterns
- **Similarity and Uncertainty**
  - Sources, Semantics,…
- **Mining Trajectories Data**
  - **Clustering**
  - **Warehousing**
- **Domain Constraints**
  - **WSN and Distributed Similarity Detection**
- **Challenges**

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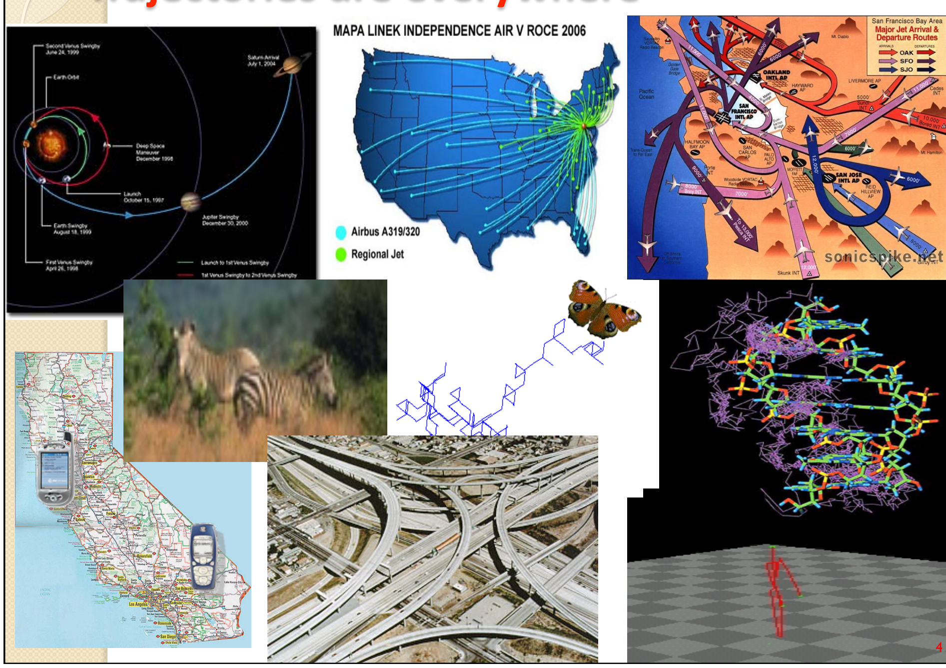
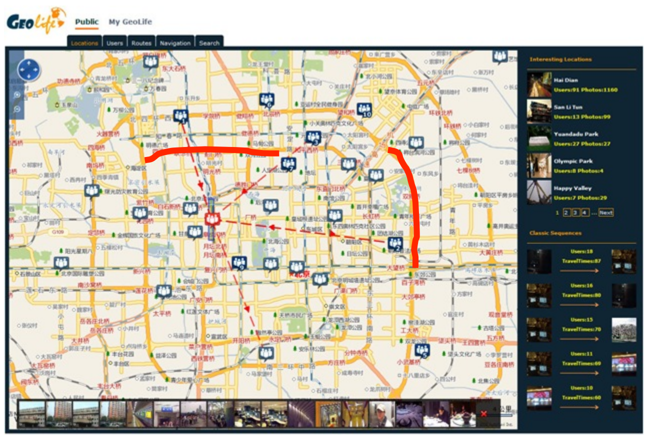
Trajectories are everywhere



Introduction/Motivation

- Plethora of Application Domains:
  - Traffic Management



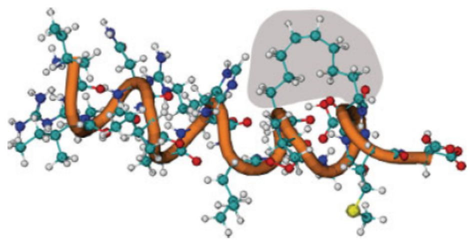
Retrieve all the ambulances which had **similar delays** patterns during the rush hours in the last 6 months

- Plethora of Application Domains:
  - Video analysis

Abnormal events detection;
Counting Pedestrians
(based on similar trajectories clustering)

[FanXW09, JiangWK07, AntoniniT06]

- Plethora of Application Domains:
  - Molecular Design

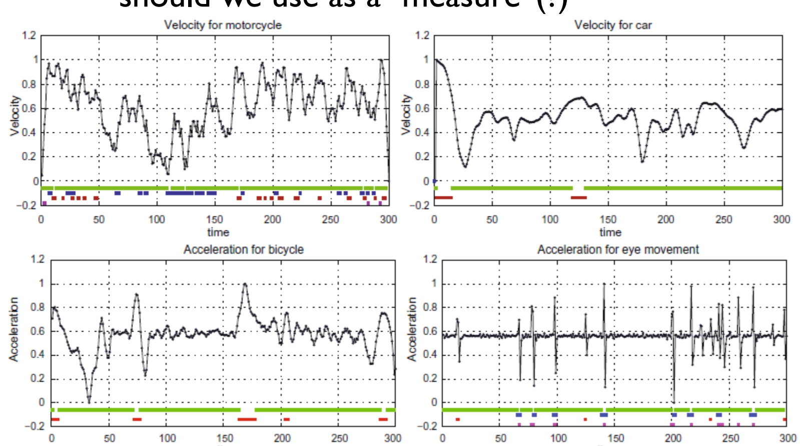


Are there any similar distributions in the molecular dynamics trajectories?

Are the motions of (sets of) cells similar?

[Hamacher09, MosigWN+09]

- Plethora of Application Domains:
  - Similarity of Motions in Different Contexts – what should we use as a "measure"(?)

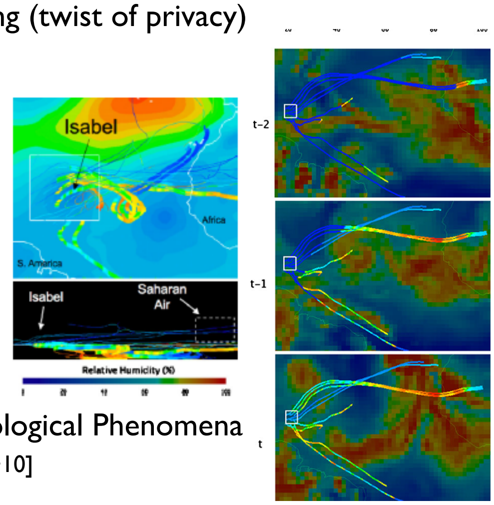


Similar Sinuosity…

Introduction/Motivation

- Plethora of Application Domains
  - Smart Phones tracking (twist of privacy)

[CLZ+11]



  - Similarity of Meteorological Phenomena
    - Humidity tracking [BBM+10]

Similarity and Distance Functions

- Fréchet distance (length of the leash)

Let C1: $[a_1,b_1] \rightarrow \mathbf{R}^2$ and C2: $[a_2,b_2] \rightarrow \mathbf{R}^2$ denote two curves their Fréchet distance is defined as

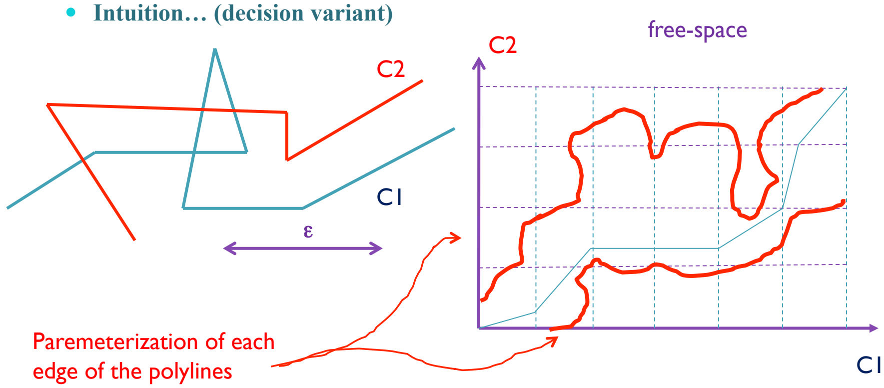
$$\delta_f(C_1,C_2) = \inf_{\alpha,\beta} \max_{t\in[0,1]} \| C_1(\alpha(t)) - C_2(\beta(t)) \|$$

α (respectively β) range over all the continuous and monotonically increasing functions $[0,1] \rightarrow [a_1,b_1]$ (respectively $[0,1] \rightarrow [a_2,b_2]$)

Clearly, a lot more general – but also harder to compute than the Hausdorff distance

Similarity and Distance Functions

- **Fréchet distance** **(length of the leash)**
  - **Intuition… (decision variant)**



$\delta_f(C_1,C_2) \leq \varepsilon$ if there is a monotonicaly increasing path

Decision: O(mn)
Computation: O(mn log(mn)) [AG95]

Note: this is for two fixed curves

Moving Objects, Trajectories

- Moving Object

Changes its spatial whereabouts over time (during its existence)

Any real-world element That may be perceived as "unique" (a car, a person)

Trajectory: continuous mapping from Time to (some) Geographical Space
$I(\subseteq) R \rightarrow R^2$

or, even think of it as parameterization over time
$\alpha(t) = (\alpha_x(t), \alpha_y(t))$

So, now
Trajectory = $\{(\alpha_x(t), \alpha_y(t), t) \mid t \in I\}$
- **Description**
- **Representation**
- **Manipulations**

[MVO+08,GBE+00]

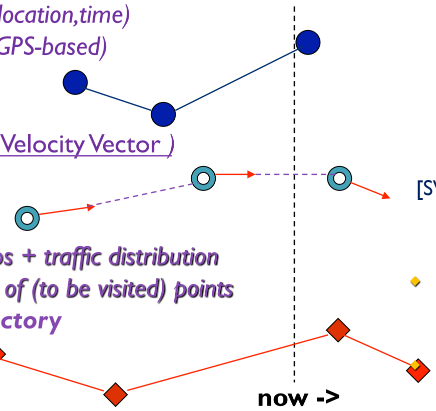
Moving Objects, Trajectories, *Time*

Future motion of moving objects can be described as:

sequence of (location,time) updates (e.g., GPS-based)

(location,time, Velocity Vector) updates

Electronic maps + traffic distribution patterns + set of (to be visited) points
=> full trajectory

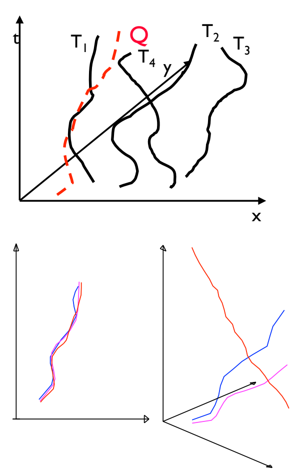


[SWC+97]

- *Transportation vehicles, patrol cars, delivery trucks, individuals travel between home and work…*
- *Over 80 million visits to online map services for trip planning*

Trajectory Similarity

- Given a query trajectory *Q* and a trajectory database *D*={T_i} search among them for the most similar trajectory
  - Determine the most similar by using not only the spatial shape of the trajectories but also their evolution in time
- Enhance the performance of the search by employing existing index structures (R-trees) used to support more traditional queries (range, NN e.t.c.).



[FGP+07] Frentzos, E., Gratsias, K., Pelekis, N., Theodoridis, Y.: Algorithms for Nearest Neighbor Search on Moving Object Trajectories. GeoInformatica 11(2): 159-193 (2007).

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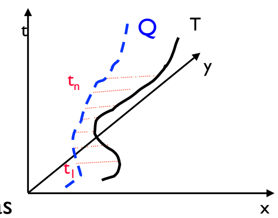
Metrics for Measuring Similarity

- Definition of Dissimilarity *DISSIM(Q,T)* between trajectories *Q* and *T* being valid during the period $[t_1, t_n]$: the definite integral of the function of time of the Euclidean distance between the two trajectories during the same period



$$DISSIM(Q,T)=\int_{t_1}^{t_n} D_{Q,T}(t)\,dt$$

- In real world, each trajectory is represented as a collection of discrete points
  1. Apply linear interpolation in between
  2. Use a discrete measure

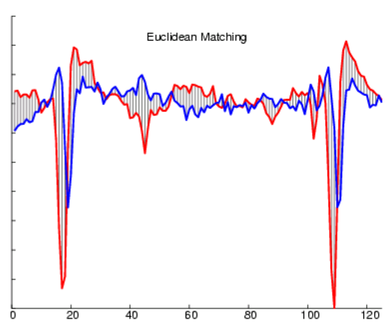
$$DISSIM(Q,T)=\sum_{k=1}^{n-1}\int_{t_k}^{t_{k+1}} D_{Q,T}(t)\,dt$$

[FGP+07]

Similarity Measures

- Euclidean (sensitive to time axis displacement)

Lp–Norm: $Lp=(\Sigma(x(i)-y(i))^p)^{1/p}$

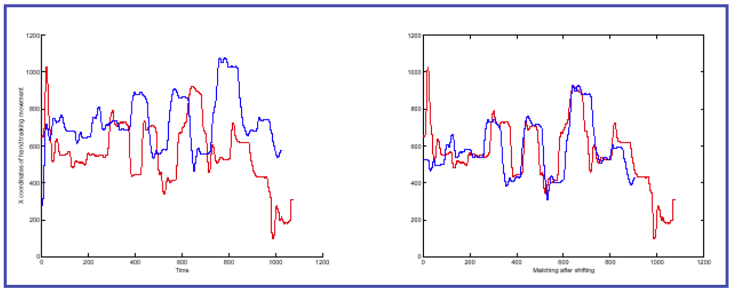


[VP06]

Requirements for Similarity Model

We need to address the following issues:

- Similar Motions in different space Regions



[VGK02] Vlachos, M., Gunopulos, D., Kollios, G.: Discovering Similar Multidimensional Trajectories. ICDE 2002

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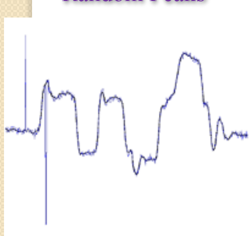
Requirements for Similarity Models

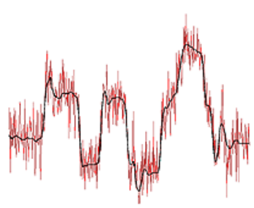
We need to address the following issues:

- Outliers

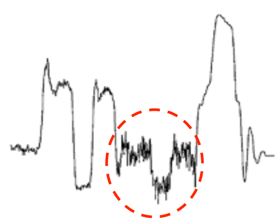
Random Peaks



Noise Everywhere



Non Recoverable Part



- Different Lengths

[VGK02]

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Dynamic time warping (DTW)

- Euclidean (L2) distance is

$$D_2(\mathbf{x}, \mathbf{y}) := \sum_{t=1}^{N} (x_t - y_t)^2$$

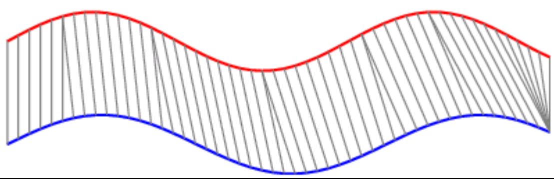
or, recursively,

$$D_2(\mathbf{x}_{1:t}, \mathbf{y}_{1:t}) = (x_t - y_t)^2 + D_2\big(\mathbf{x}_{1:(t-1)}, \mathbf{y}_{1:(t-1)}\big).$$

- Dynamic time warping distance is

$$D_w(\mathbf{x}_{1:i}, \mathbf{y}_{1:j}) := (x_i - x_j)^2 + \min \begin{cases} D_w\big(\mathbf{x}_{1:(i-1)}, \mathbf{y}_{1:(j-1)}\big) & \\ D_w\big(\mathbf{x}_{1:(i-1)}, \mathbf{y}_{1:j}\big) & \leftarrow \text{shrink } \mathbf{x} \text{ / stretch } \mathbf{y} \\ D_w\big(\mathbf{x}_{1:i}, \mathbf{y}_{1:(j-1)}\big) & \leftarrow \text{stretch } \mathbf{x} \text{ / shrink } \mathbf{y} \end{cases}$$

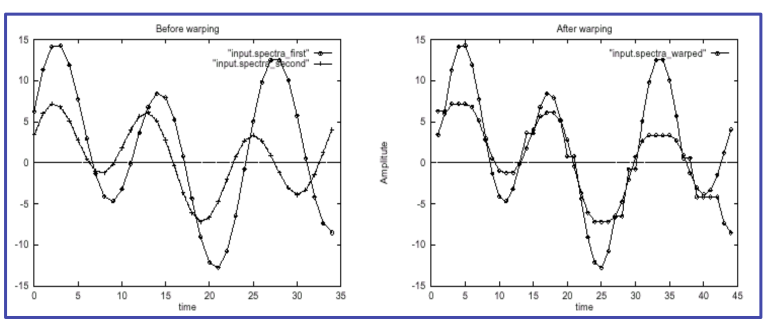
where $\mathrm{x}_{1:i}$ is the subsequence $(x_1, \ldots, x_i)$



[VP06]

Dynamic Time Warping (DTW)

- **Time Warping**
  - **Allows stretching in time axis**
  - **Matches all elements**
  - **Extensively used in speech recognition domain**



[VGK02]

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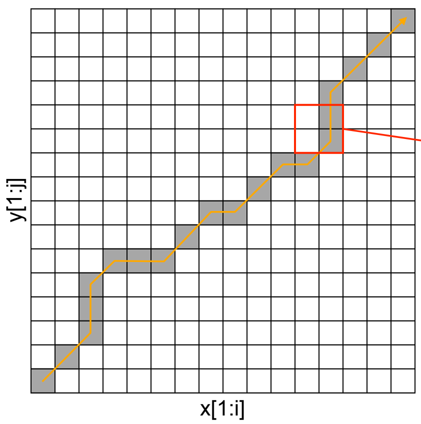
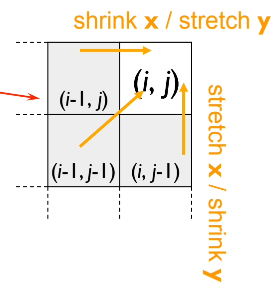
Dynamic time warping algorithm

- DTW allows any path
- Examine all paths:





- Standard dynamic programming to fill in table—top-right cell contains final result

[VP06] Vlachos, M., Papadimitriou, S.:Hands-on Time-Series Analysis with Matlab. Tutorial, ICDM 2006

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Longest Common Subsequence (LCSS)

Arithmetic Example:

- **t1=[0, 4, 6, 8, 7, 4, 6, 5, 6, 4, 6]**
- **t2=[0, 3, 4, 6, 7, 6, 3, 6, 4, 6]**

- **Definition**

$$LCSS(A,B) = \begin{cases} 1 + LCSS(Head(A), Head(B)), & \text{if } a_n = b_n \\ max(LCSS(Head(A),B), LCSS(A, Head(B))), & \text{otherwise} \end{cases}$$

- **Dynamic Programming Solution**

$$LCSS[i,j] = \begin{cases} 0, & \text{if } i = 0 \\ 0, & \text{if } j = 0 \\ 1 + LCSS[i-1, j-1], & \text{if } a_i = b_j \\ max(LCSS[i-1, j], LCSS[i, j-1]), & \text{otherwise} \end{cases}$$

[VGK02]

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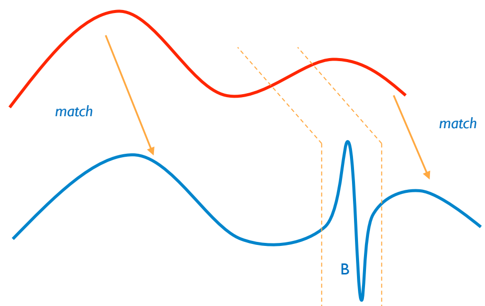
Differences between DTW & LCSS

- DTW has to much every element, so is more sensitive to outliers
  LCSS matches only the similar parts

[VGK02]

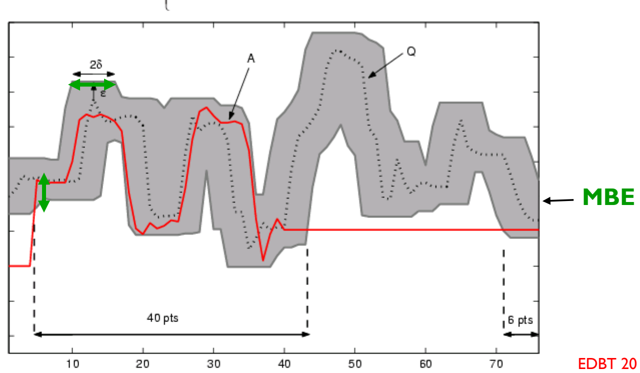


[DTSWK08] provides experimental comparison of 8 representation methods; 9 similarity measures (+variants); 38 datasets. [WMDTSK12] provides further extensions.

LCSS for real values

$$LCSS_{\delta,\varepsilon}(A,B)=\begin{cases}0 & \textit{if } \text{A or B is empty}\\ 1+LCSS_{\delta,\varepsilon}(Head(A),Head(B)) & \\ \quad \textit{if } |a_n-b_n|<\varepsilon \textit{ and } |n-m|\le\delta & \\ \max(LCSS_{\delta,\varepsilon}(Head(A),B),LCSS_{\delta,\varepsilon}(A,Head(B))) & \\ \quad \textit{otherwise} & \end{cases}$$

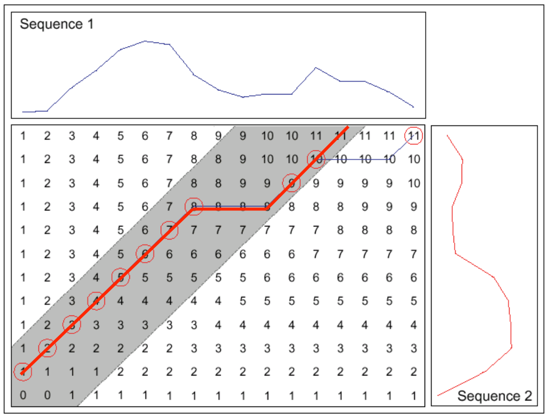
[VGK02]



LCSS Dynamic Programming algorithm

LCSS can be computed in O(δ (l_1+l_2)) by dynamic programming algorithm.

$$LCSS[i,j] = \begin{cases} 0 & \text{if } i = 0 \\ 0 & \text{if } j = 0 \\ 1 + LCSS[i-1, j-1] & \text{if } a_i = b_i \\ \max(LCSS[i-1, j], LCSS[i, j-1]) & \text{otherwise} \end{cases}$$



$$S1(\delta, \varepsilon, A, B) = \frac{LCSS_{\delta,\varepsilon}(A,B)}{min(n,m)}$$

[VGK
Multi

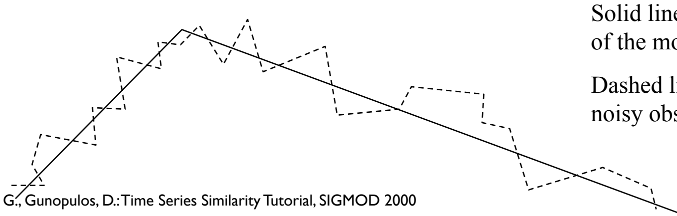
Other approaches: Probabilistic Generative Modeling Method (Ge et al, 2000)

- Given sequence Q, construct a model M_Q(i.e. a probability distribution on waveforms)
- Given a new pattern Q', measure similarity by computing p(Q'|M_Q)

Markov Model Example:

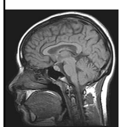
Solid lines: the two states of the model

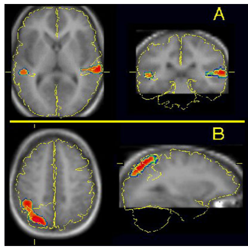
Dashed lines: the actual noisy observations

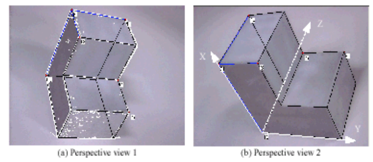


[DG00] Das. G., Gunopulos, D.: Time Series Similarity Tutorial, SIGMOD 2000

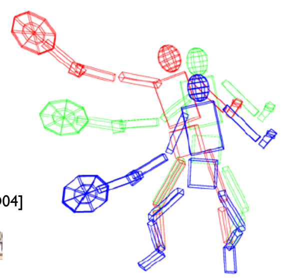
Rotation Invariant Similarity







- Robot vision (object contour recognition)
- Medical applications (CAT scan registration)
- Handwriting recognition (tablet PCs, signatures)
- Pose estimate –MOCAP data (video games, movies)



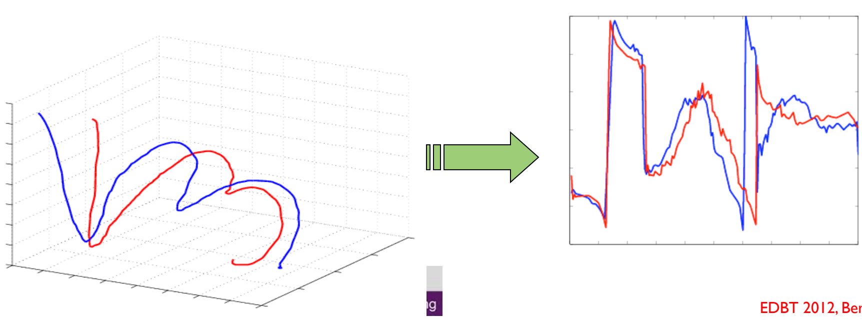
[VGD04]

Space Transformation

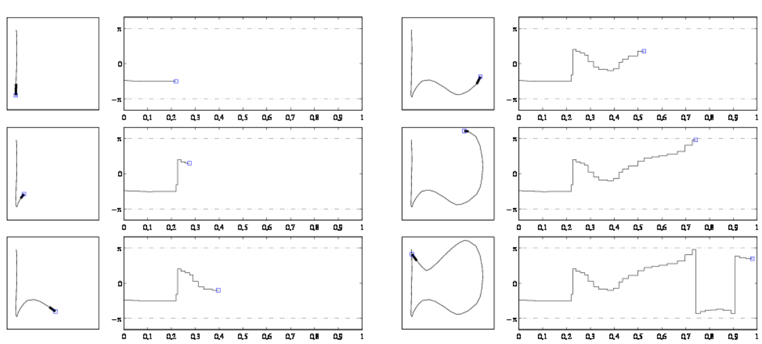
Method:

- Transform trajectory to rotation invariant space
  - *Positional information -> Angle/ Arc length information*
- Proper normalization ensures also:
  - *Translation invariance*
  - *Scale Invariance*
- Perform matching in new space
  - *Use Warped Matching to compensate for shape variations*

[VGD04] Vlachos, M., Gunopulos, D., Das, G.: Rotation invariant distance measures for trajectories. KDD 2004



Mapping Example



Angles to utilize:

- Exact angles (*as in figure*)
- Relative Angles (to previous segment)
- Angles from center of mass

[VGD04]

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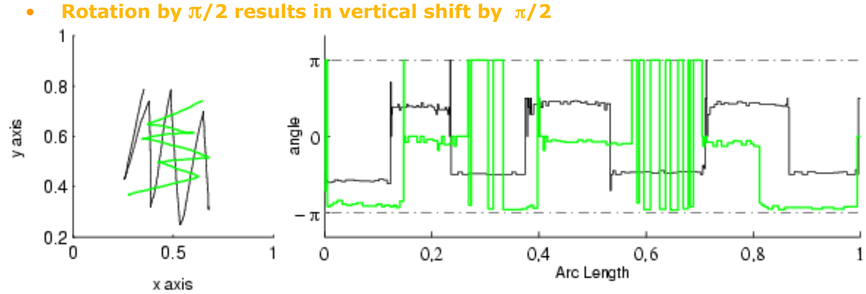
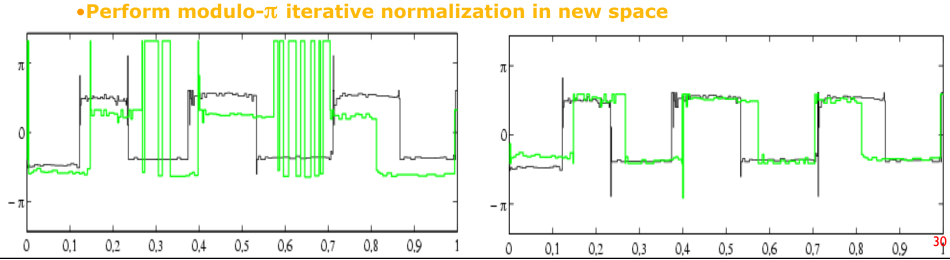
Mapping Refinement (Vlachos et al 04)

- **Rotation by $\pi/2$ results in vertical shift by $\pi/2$**



- **Perform modulo-π iterative normalization in new space**



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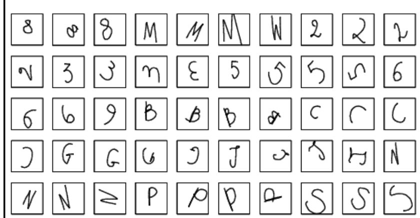
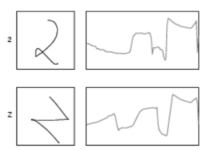
Handwriting recognition example





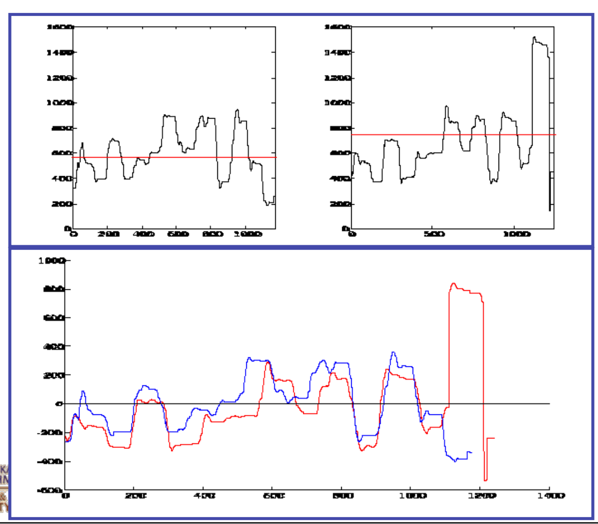
[VGD04] *"Bad" Classifications*

2D Mapping according to new distance.

Rotated versions are mapped close in space

Parallel Movements: Normalization

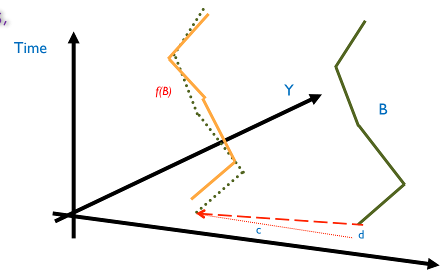
- Normalization (average value subtraction) does not guarantee best match for LCSS or DTW



[VGK02]

Extending LCSS: Translation Invariant Similarity

- Standard techniques cannot detect parallel movements,



- So, we define S2:

$$\boxed{S2 = \max_{f_{c,d}\in F} S1(\delta,\varepsilon,f_{c,d}(B))}$$

- S2 *can* detect parallel movements
- Better accuracy than simple normalization
- Distance *D1= 1-S1* & distance *D2 = 1-S2*

[VGK02]

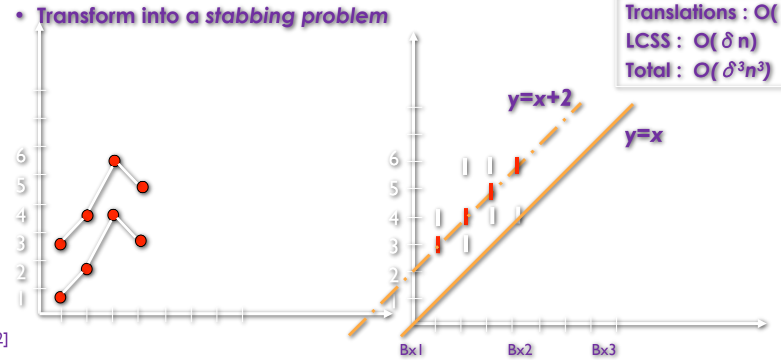
Exact Algorithm for Translation Invariant Similarity

For trajectories A, B with length n we want to find:

- translation $f_{c,d}$ that maximizes LCSS between A and $f_{c,d}(B)$
- Not infinite translations.
  - Each dimension separately
  - A translation in 1D: *fc(bi) = bi + c* (line with slope 1)
  - *fc(bi)* will allow bi to be matched to all aj: $|i\text{-}j| < \delta$ & $ai\text{-}\varepsilon \le fc(bi) \le (bi, aj+\varepsilon)$

- ***Transform into a stabbing problem***

Translations : $O(\delta^2 n^2)$
LCSS : $O(\delta n)$
Total : $O(\delta^3 n^3)$

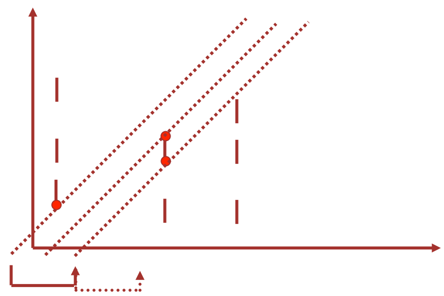


[VGK02]

Approximate Algorithm

A translation corresponds to a line *fc(x) = x+c*.

- **Sort translations by c → THEY DIFFER IN HOW MANY SEGMENTS?**



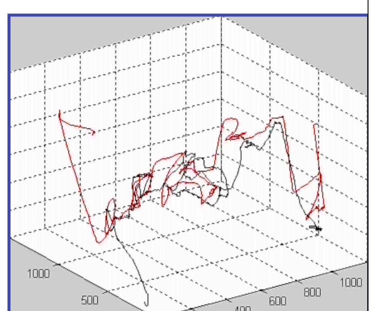
- **If we can afford to be within β of *max(Sim)* → we can afford to lose *β n* elements**
- Don't take all translations → we can examine every βn translations each time
- So, if we examine every βn, we lose *at most* βn elements (1D)
- So, for 2D, we can skip every βn/2 translations

[VGK02]

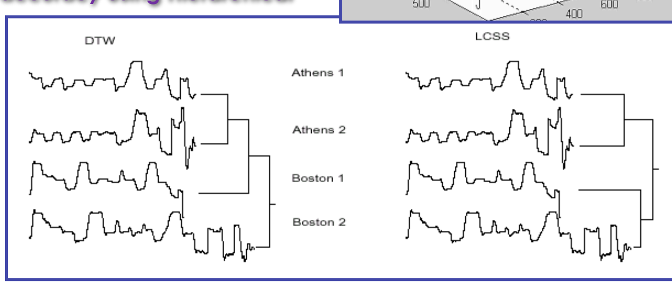
Classification Accuracy

Example using Video Tracking data

- **The 2D time series of a human tracking feature (e.g. the end of a finger). In conjunction with a "spelling program" the user can "write" various words.**
- **The data we used correspond to: *'athens', 'berlin', 'london', 'boston', 'paris'*.**



Test classification accuracy using Hierarchical Clustering



[VGK02]

Indexing using the Triangle Inequality

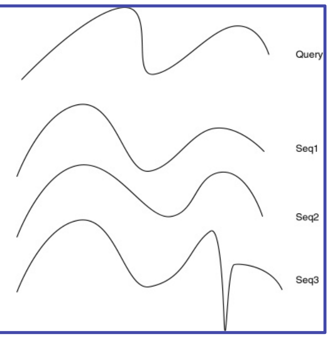
- *Objective: Create an index to answer Nearest Neighbor Queries for new trajectories*

Pruning power of triangle inequality:

- best-match-so-far = 20, D(Q,S2) = 150.
- D(S2,S1) = 20 and D(Q,S1) >= D(Q,S2)-D(S2,S1) => D(Q,S1) >= 150 – 20 = 130 we can prune S1.
- Similarly, we can prune S3.

Pairwise Distances

	Seq1	Seq2	Seq3
Seq1	0	20	110
Seq2	20	0	90
Seq3	110	90	0



[AFS93][VGK02]

Metric Properties of Similarity Measures

Triangle inequality does not hold for DTW or LCSS

(but holds for Edit Distance or DTW with gaps).

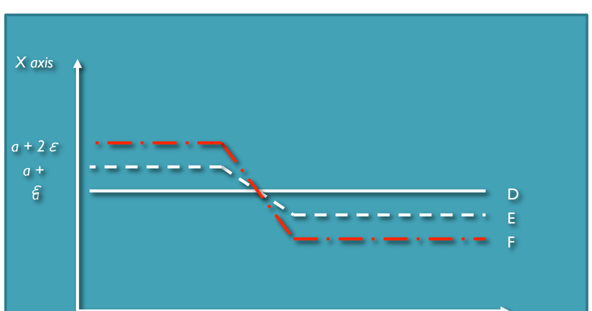
Example 1: A = 1,1,5

B = 2,2,5

C = 2,2,2,2,2,5

Then: DTW(A,B) = 2, DTW(B,C) = 0, DTW(A,C) = 5

Example 2:



D1(δ, ε, A, C) > D2(δ, ε, A, B) + D2(δ, ε, B, C), where D2(δ, ε, A, B)=1-LCSS(δ, ε, A, B).

ΕΘΝΙΚΟΝ & ΚΑΠΟΔΙΣΤΡΙΑΚΟΝ ΠΑΝΕΠΙΣΤΗΜΙΟΝ ΑΘΗΝΩΝ
NATIONAL & KAPODISTRIAN UNIVERSITY OF ATHENS

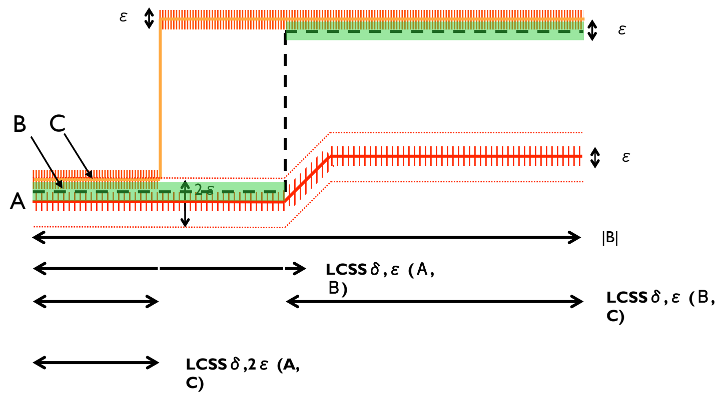
McCormick Northwestern Engineering

Quasi triangle inequality for LCSS

- **Lemma 1:** *Given trajectories A, B, C,*
  $LCSS_{\delta,2\varepsilon,F}(A, C) \geq LCSS_{\delta,\varepsilon,F}(A, B) + LCSS_{\delta,\varepsilon,F}(B, C) - |B|$

|B| is the length of the seq. B.

- **Proof:** $LCSS_{\delta,2\varepsilon,F}(A, C)$ = (#B points that match A and C within ε) $\geq$ (#points in B) – (#B points that don't match A) - (#B points that don't match C)



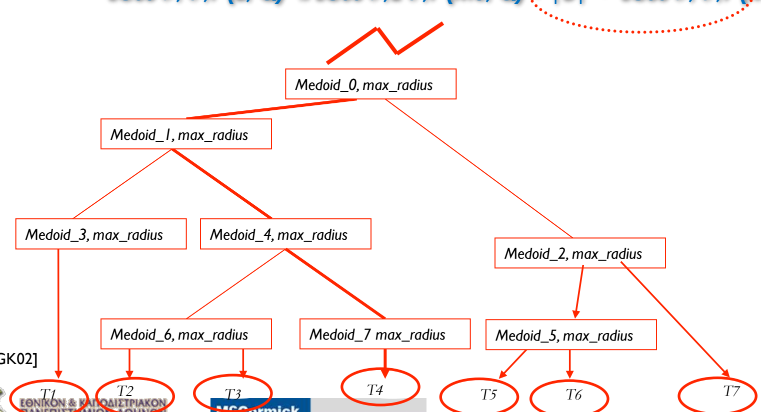
[VGK02]

Creating an indexing structure

- **Index:** create hierarchical clustering tree (using pairwise distances). Also, also keep **Medoid** of cluster and **radius** of cluster from medoid

radius

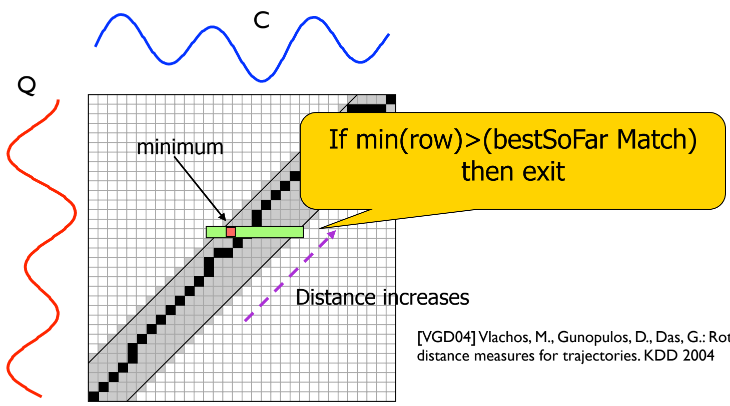
- **Prune sub-trees using the previous inequality:**
  $LCSS_{\delta,\varepsilon,F}(B, Q) \leq LCSS_{\delta,2\varepsilon,F}(Mc, Q) + |B| - LCSS_{\delta,\varepsilon,F}(Mc, B)$



[VGK02]

Sequential Scan

- Compare the Query to all the trajectories in the database
- Use Lower/Upper Bounds or Early Termination to speed up the search:



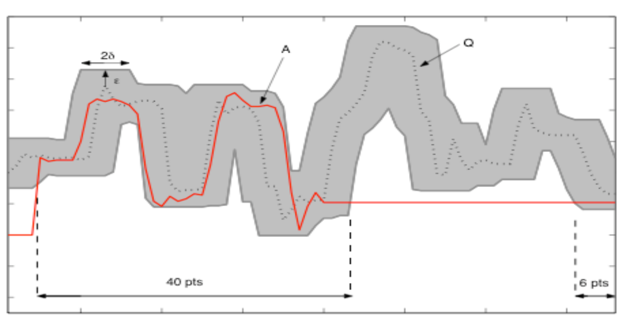
[VGD04] Vlachos, M., Gunopulos, D., Das, G.: Rotation invariant distance measures for trajectories. KDD 2004

Computing Bounds: The Envelope Approach

$$EnvLow \leq MBE_{\delta,\varepsilon}(Q) \leq EnvHigh \text{ where } \begin{cases} EnvHigh[i] = \max\{Q[j]+\varepsilon\}, & |i-j|<\delta \\ EnvLow[i] = \min\{Q[j]-\varepsilon\}, & |i-j|<\delta \end{cases}$$

$$LCSS_{\delta,\varepsilon}(MBE(Q), A) = \sum \begin{cases} 1, & \text{if A[i] within envelop} \\ 0, & \text{therwise} \end{cases}$$

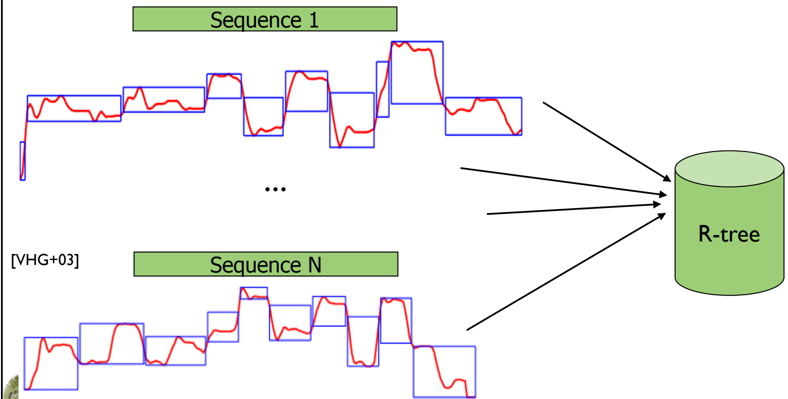
Theorem: $LCSS_{\delta,\varepsilon}(Q, A) \leq LCSS_{\delta,\varepsilon}(MBE(Q), A)$



[VHG+03] Vlachos, M., Hadjieleftheriou, M., Gunopulos, D., Keogh, E.J.: Indexing multi-dimensional time-series with support for multiple distance measures. KDD 2003

Envelope Representation, Dimensionality Reduction (Vlachos et al 03)

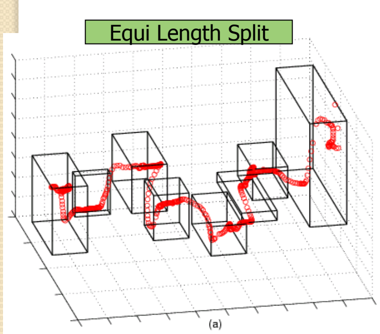
- MBRs – allow tight coupling with index structure

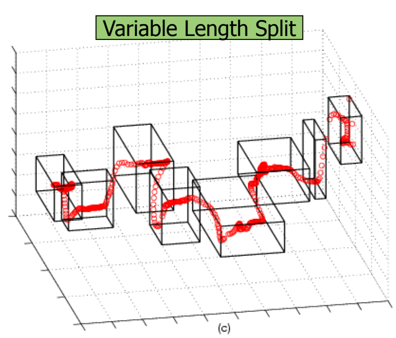


[VHG+03]

MBR Representation

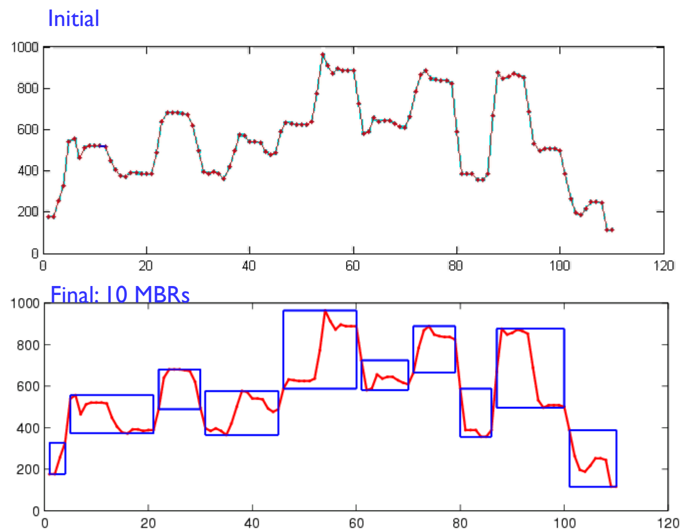
- Approximate Trajectories with 3D MBRs
  - Equi-Split (*O(n)*)
  - Optimal (*dynamic programming, O(n^2k)*)
  - Greedy Merge (*O(nlogn)*)





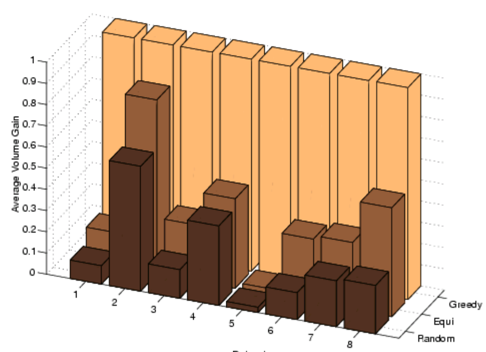
[VHG+03]

Greedy Algorithm



[VHG+03]

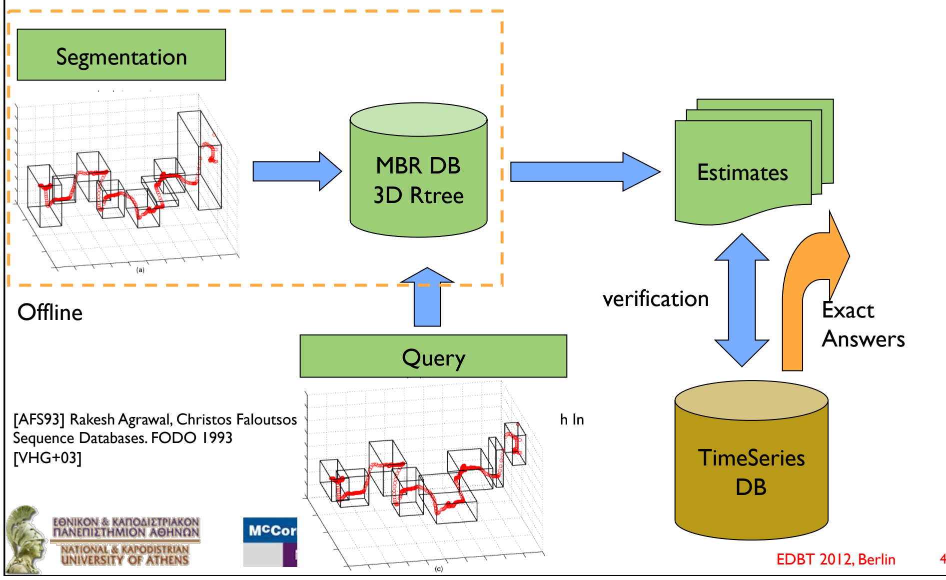
MBR generation Comparison



[VHG+03]

- Equi-Split slightly better than random

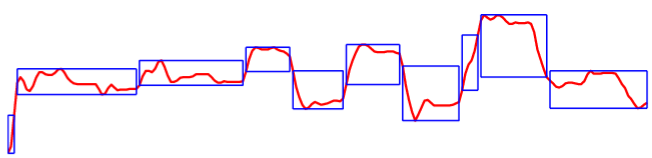
Filtering using Bounds on the Similarity



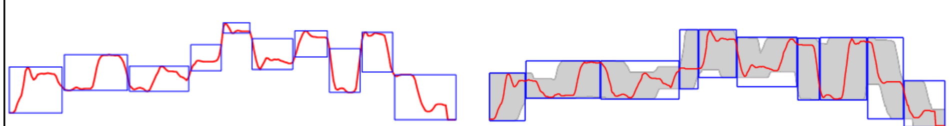
[AFS93] Rakesh Agrawal, Christos Faloutsos ... h In
Sequence Databases. FODO 1993
[VHG+03]

One Index for all?

- Common Representation for Indexed Sequences



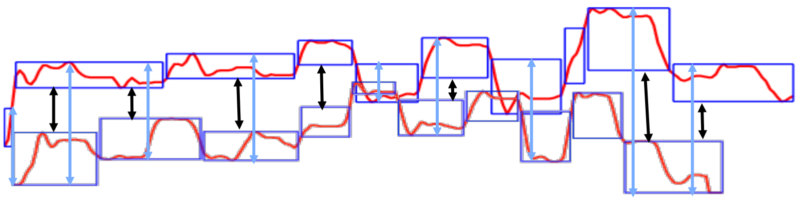
- Different Representation for Queries



Euclidean Distance | DTW or LCSS

[VHG+03] Vlachos, M., Hadjieleftheriou, M., Gunopulos, D., Keogh, E.J.: Indexing multi-dimensional time-series with support for multiple distance measures. KDD 2003

Euclidean Lower Bound



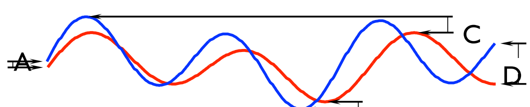
Lower Bounds

Upper Bounds

[VHG+03]

Lower bounds for DTW

LB_Kim

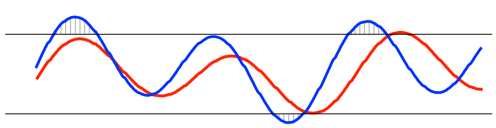


Kim, S, Park, S, & Chu, W. *An index-based approach for similarity search supporting time warping in large sequence databases.* ICDE 01, pp 607-614

LB_Yi

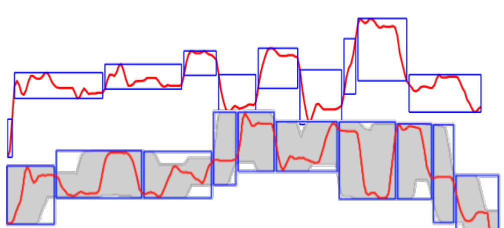
max(Q)

min(Q)



Yi, B, Jagadish, H & Faloutsos, C. *Efficient retrieval of similar time sequences under time warping.* ICDE 98, pp 23-27

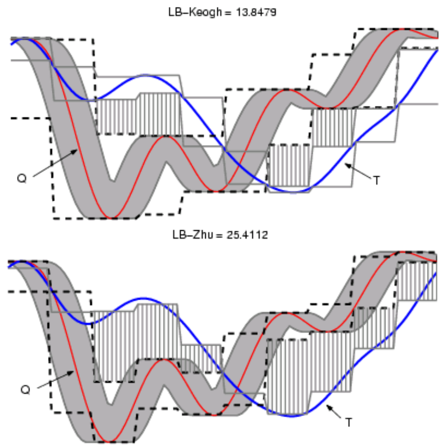
Using MBRs (Vlachos et al. 2003)



[VHG+03]

Tightness of approximation

- **Prune most of sequences using linear cost "Lower Bounds"**
  - *Envelope approaches*



[VHG+03]

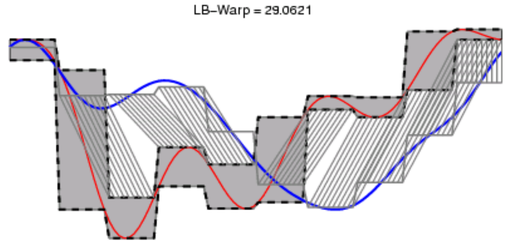
Tightness of Approximation (Vlachos et at 04)

- **Perform direct warping between approximations**
  - *Calculate PAA(T)*
  - *Create MBR(Q)*

$$D_{seg}(MBR_i(Q), PAA_j(T)) = \begin{cases} k \times \Delta(\hat{q}_{i,L}, \bar{t}_j), & \text{if} \quad \hat{q}_{i,L} > \bar{t}_j \\ k \times \Delta(\hat{q}_{i,U}, \bar{t}_j) & \text{if} \quad \hat{q}_{i,U} < \bar{t}_j \\ 0 & \text{otherwise} \end{cases}$$

Lemma: $LB - Warp_{\lceil \delta(\frac{k}{n}) \rceil}(MBR(Q), PAA(T)) \leq DTW_\delta(Q,T)$

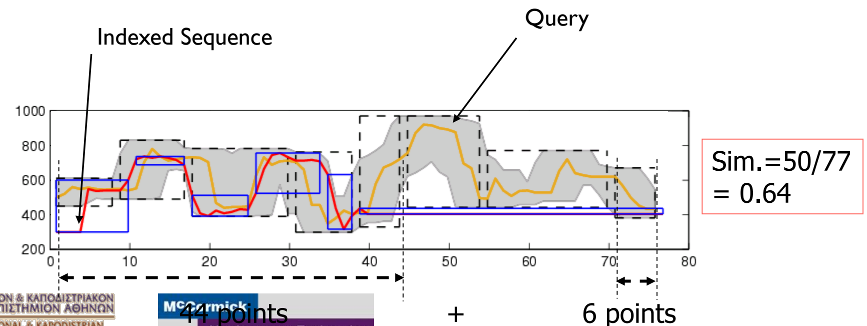
- **New Lower Bounds Closer to original warped distance**



[VGD04]

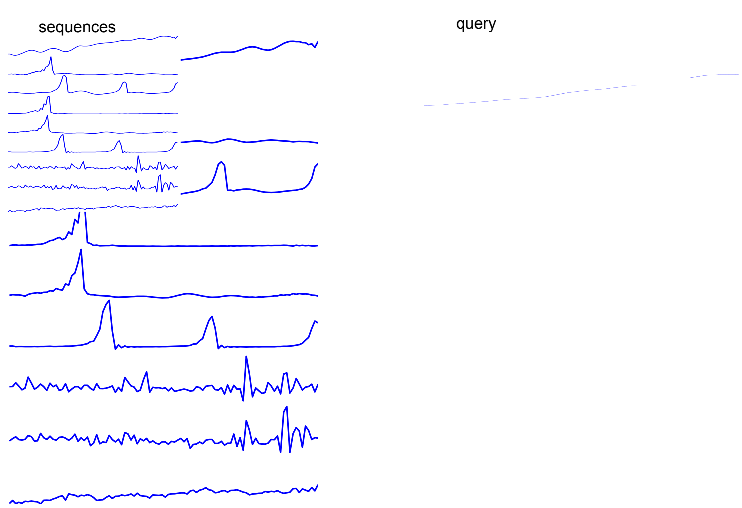
LCSS Upper Bound (Vlachos et al 03)

- LCSS captures similarity
- Similarity = 1 - dist



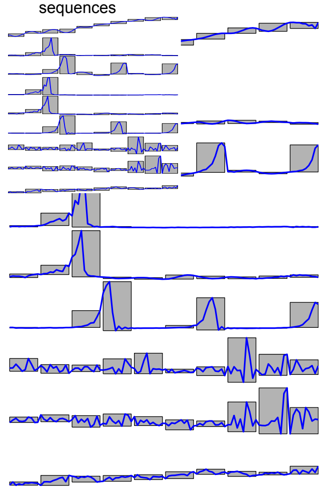
[VHG+03]

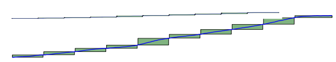
Using bounds for filtering: Example



[VHG+03]

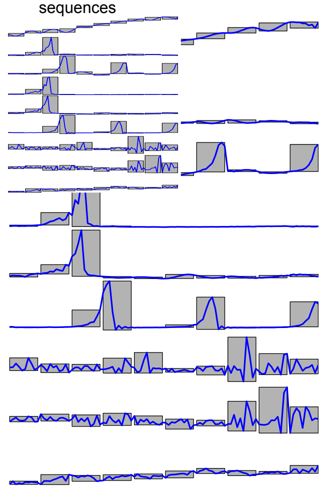
Using bounds for filtering: Example





[VHG+03]

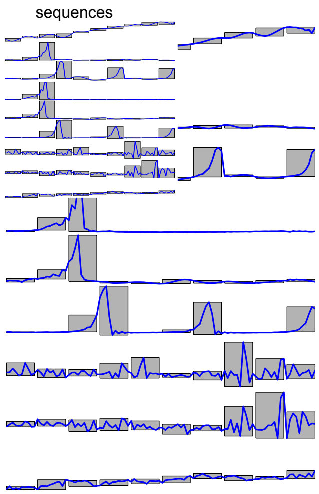
Using bounds for filtering: Example



Lower Bounds	Upper Bounds
4.6399	126.1381
37.9032	259.7098
19.5174	291.3299
72.1846	198.9709
67.1436	208.0821
78.0920	190.4292
70.9273	187.1943
63.7253	201.0270
1.4121	46.6011

[VHG+03]

Using bounds for filtering: Example

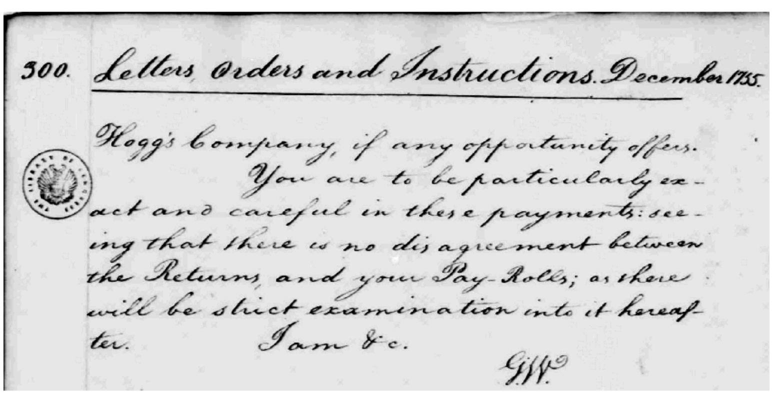


Lower Bounds	Upper Bounds	True Distance	
4.6399	126.1381	46.7790	
37.9032	259.7098	108.8856	
19.5174	291.3299	113.5873	
~~72.1846~~	198.9709	104.5062	
~~67.1436~~	208.0821	119.4087	
~~78.8920~~	190.4292	120.0066	
~~70.9273~~	187.1943	111.6011	
~~63.7253~~	201.0270	119.0635	
1.4121	46.6011	17.2540	BestSoFar

[VHG+03]

Application to Image Data

- Library of Congress has 54 million manuscripts (20TB of text)

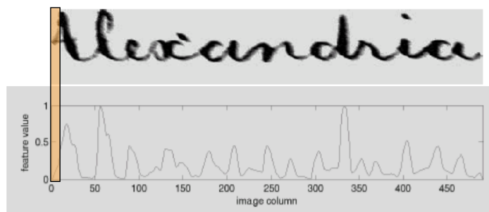

300. Letters, Orders and Instructions. December 1755.

Hogg's Company, if any opportunity offers.
You are to be particularly ex-
act and careful in these payments: see-
ing that there is no disagreement between
the Returns, and your Pay-Rolls; as there
will be strict examination into it hereaf-
ter. I am &c.
G W.

[VHG+03]

Application to Image Data

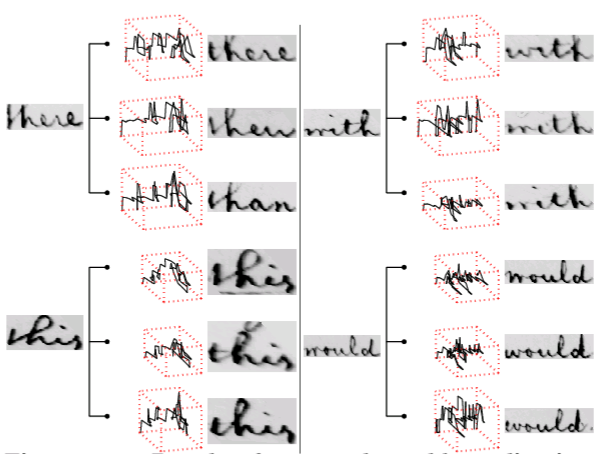
- Extract words from document
- Extract image features from words
- Annotate a subset of the words
- Classify the rest of the document words



Word Image Matching Using Dynamic Time Warping, Toni M. Rath and R. Manmatha

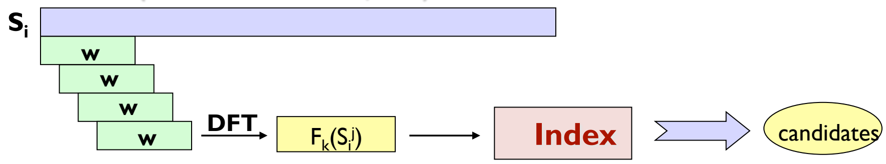
[VHG+03]

Application to Image Data (cont/d)

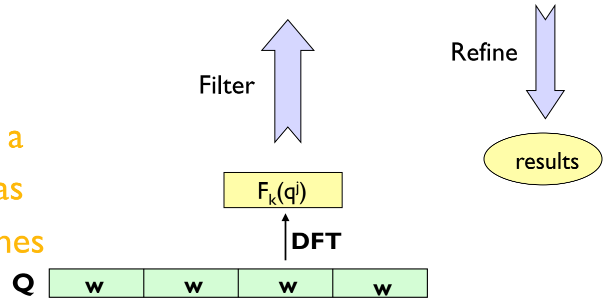


[VHG+03]

ST-Index: Subsequence Matching Using the Euclidean Distance (Faloutsos et al, 94)



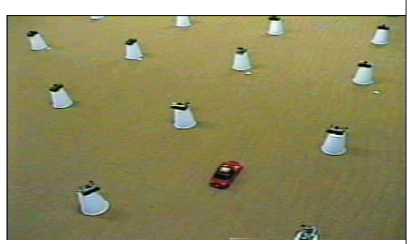
- The Problem:
  Find if there exists a sequence Si that has a region that matches with Q.



[FRM94] Christos Faloutsos, M. Ranganathan, Yannis Manolopoulos: Fast Subsequence Matching in Time-Series Databases. SIGMOD Conference 1994

Trajectory Similarity Search in a Distributed Environment

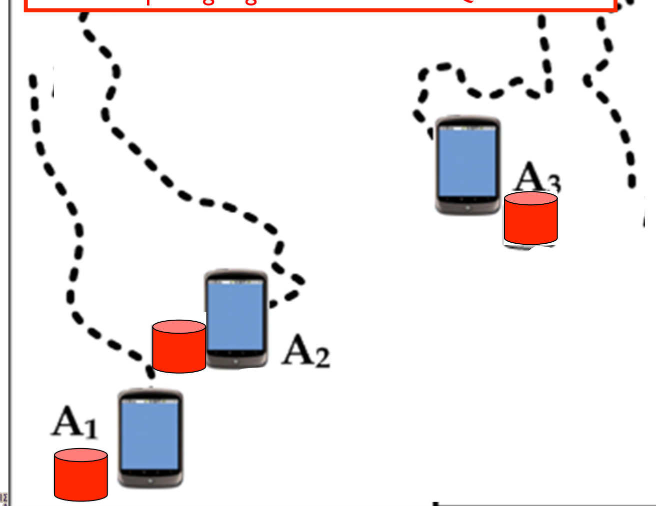
- Habitat monitoring
  - Animal migration patterns
- Surveillance video
- Camera sensor network
  - Each sensor can monitor the movement of objects within a small area



Sensor Network constraints ->
    Minimize communication costs
Use In-situ storage, In-network processing

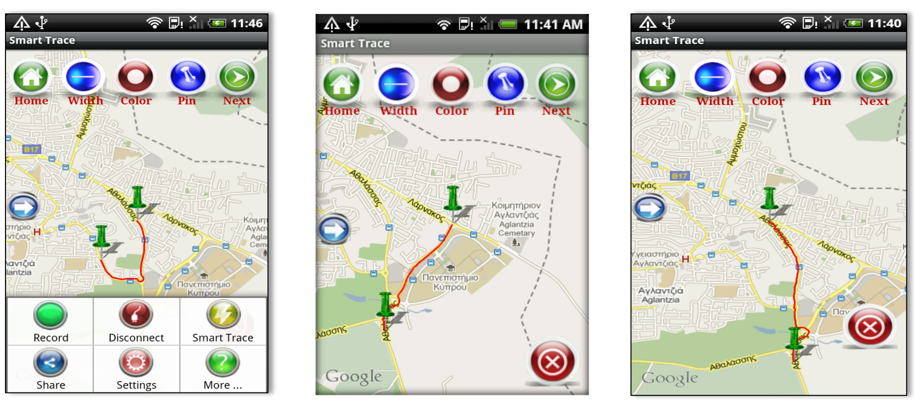
SmartTrace: Fully distributed, privacy preserving trajectory search application using mobile phones

Find the K most similar trajectories to Q without pulling together all traces at QN



SmartTrace

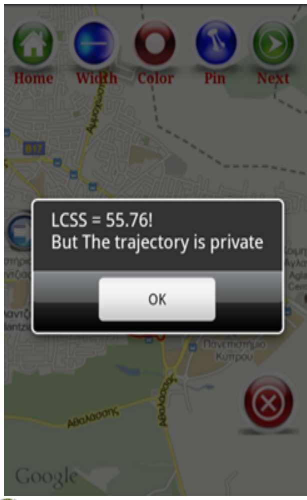
- Server implemented in JAVA (4,500 LOC)
- Client implemented in JAVA on Android (2,500 LOC + XML files)



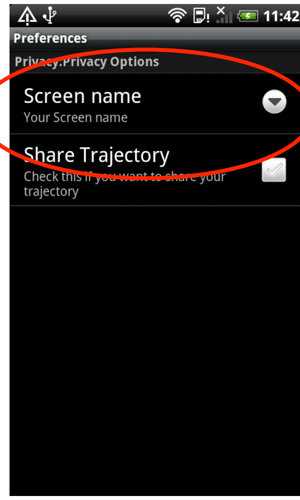
Query Device B Device C

[CLZ+11]

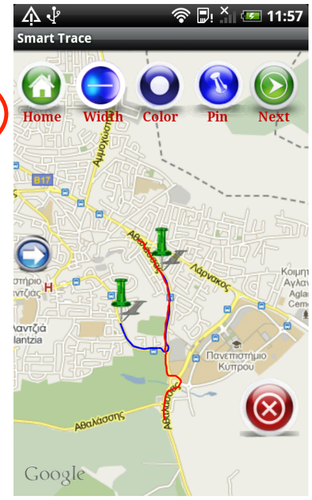
SmartTrace



Answer



Privacy Setting



Answer With Trace